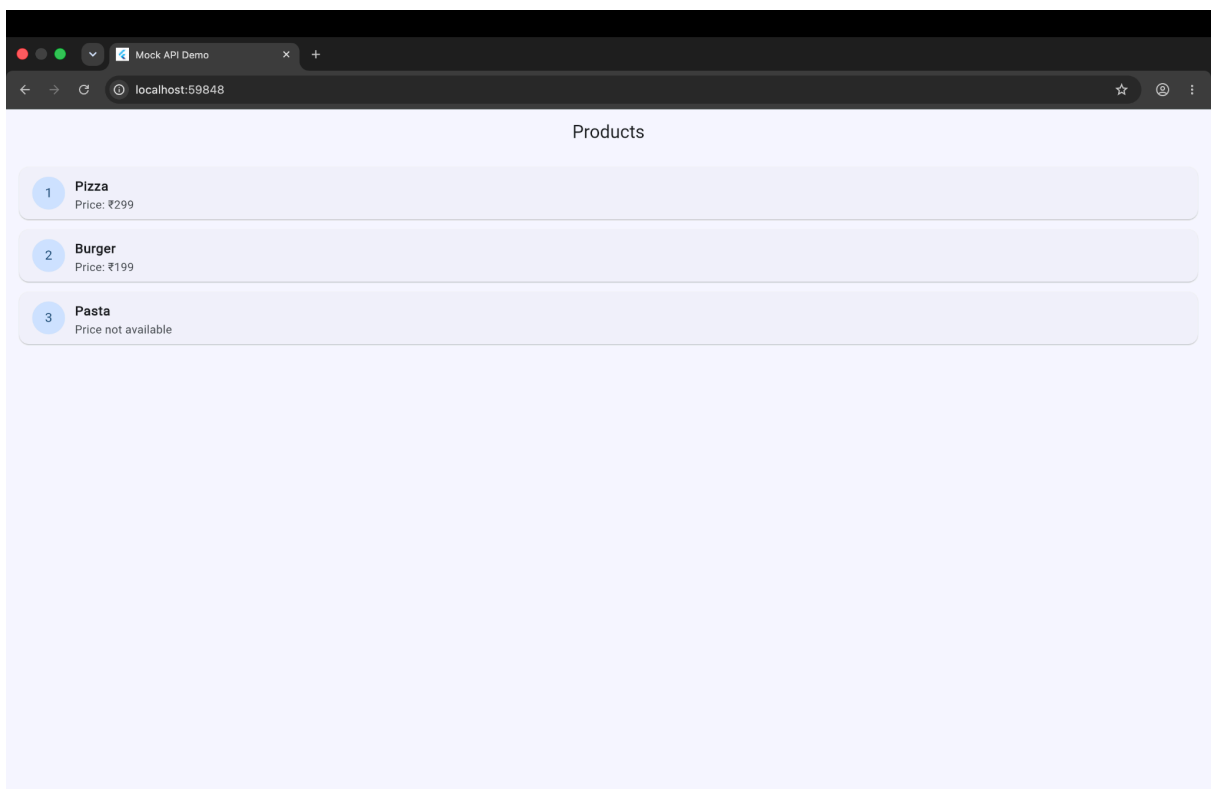
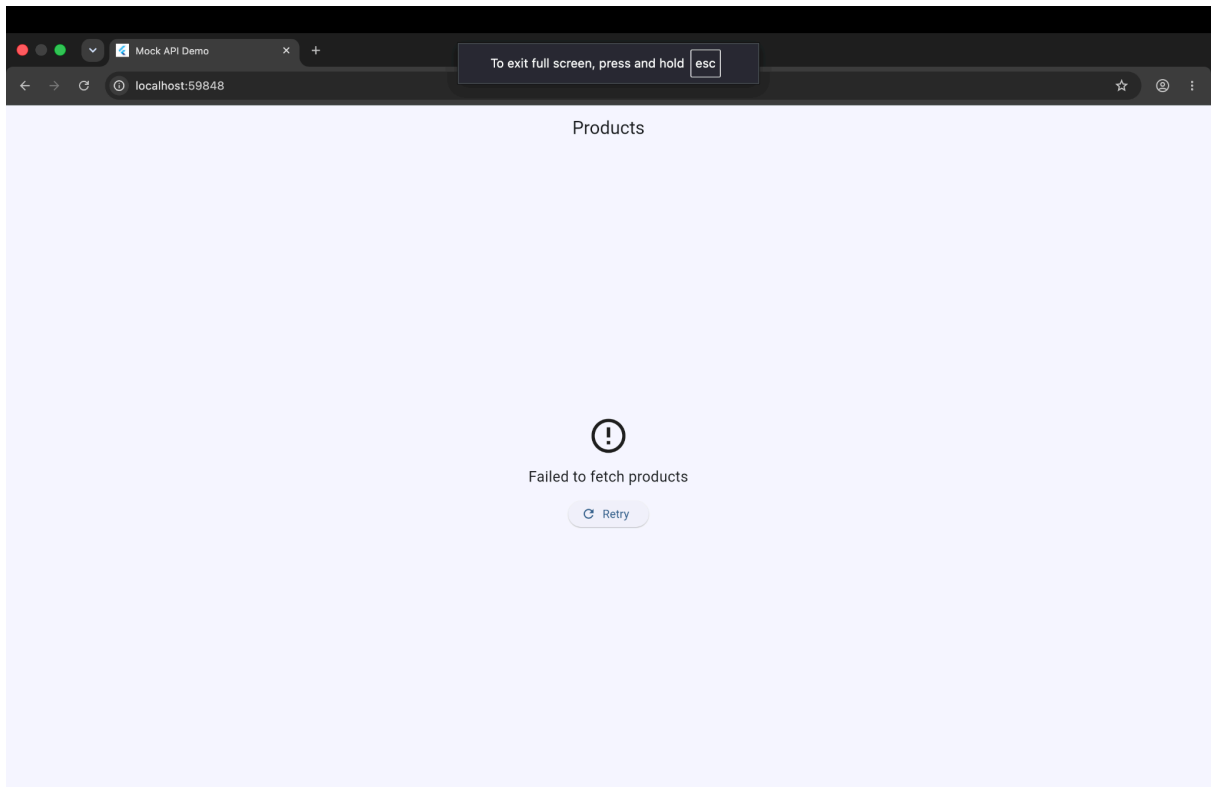




Main.dart

```
import 'package:flutter/material.dart';  
  
import 'mock_api.dart';
```

```
import 'product.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Mock API Demo',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.blue),
        useMaterial3: true,
      ),
      home: const ProductPage(),
    );
  }
}

class ProductPage extends StatefulWidget {
  const ProductPage({super.key});

  @override
  State<ProductPage> createState() => _ProductPageState();
}

class _ProductPageState extends State<ProductPage> {
  final MockApi api = MockApi();

  List<Product>? products;
  String? errorMessage;
  bool isLoading = true;

  @override
  void initState() {
    super.initState();
    fetchData();
  }

  Future<void> fetchData() async {
    setState(() {
```

```

        isLoading = true;
        errorMessage = null;
    });

    try {
        final result = await api.fetchProducts();

        setState(() {
            products = result;
            isLoading = false;
            errorMessage = null;
        });
    } catch (error) {
        setState(() {
            errorMessage = 'Failed to fetch products';
            isLoading = false;
        });
    }
}

@override
Widget build(BuildContext context) {
    return Scaffold(
        appBar: AppBar(title: const Text('Products')),
        body: buildBody(),
    );
}

Widget buildBody() {
    // Loading state
    if (isLoading) {
        return const Center(child: CircularProgressIndicator());
    }

    // Error state
    if (errorMessage != null) {
        return Center(
            child: Column(
                mainAxisAlignment: MainAxisAlignment.center,
                children: [
                    const Icon(Icons.error_outline, size: 50),
                    const SizedBox(height: 12),
                    Text(
                        errorMessage!,
                        style: const TextStyle(fontSize: 18),

```

```

        textAlign: TextAlign.center,
      ),
      const SizedBox(height: 16),
      ElevatedButton.icon(
        onPressed: fetchData,
        icon: const Icon(Icons.refresh),
        label: const Text('Retry'),
      ),
    ],
  ),
);
}

// Null or empty response
if (products == null || products!.isEmpty) {
  return const Center(
    child: Text('No products available', style: TextStyle(fontSize: 18)),
  );
}

// Display API data
return ListView.builder(
  padding: const EdgeInsets.all(8),
  itemCount: products!.length,
  itemBuilder: (context, index) {
    final product = products![index];

    return Card(
      margin: const EdgeInsets.symmetric(horizontal: 8, vertical: 6),
      child: ListTile(
        leading: CircleAvatar(child: Text('${product.id}')),
        title: Text(
          product.name,
          style: const TextStyle(fontWeight: FontWeight.bold),
        ),
        subtitle: Text(
          product.price == null
            ? 'Price not available'
            : 'Price: ₹${product.price}',
        ),
      ),
    );
  },
);
}

```

```
}
```

Product.dart

```
class Product {  
  final int id;  
  final String name;  
  final double? price;  
  
  Product({  
    required this.id,  
    required this.name,  
    this.price,  
  });  
}
```

Mock_api.dart

```
import 'product.dart';  
  
class MockApi {  
  // Simulates an error on the first API request.  
  bool _firstRequest = true;  
  
  Future<List<Product>?> fetchProducts({bool returnNull = false}) async {  
    // Simulate network delay.  
    await Future.delayed(const Duration(seconds: 2));  
  
    // Simulate API failure on the first request.  
    if (_firstRequest) {  
      _firstRequest = false;  
      throw Exception('Failed to fetch products');  
    }  
  
    // Simulate a null API response.  
    if (returnNull) {  
      return null;  
    }  
  
    // Successful API response.  
    return [  
      Product(id: 1, name: 'Pizza', price: 299),  
      Product(id: 2, name: 'Burger', price: 199),  
    ]  
  }  
}
```

```
    Product(id: 3, name: 'Pasta', price: null),  
  ];  
}  
}
```